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FELICS: Design, Implementation, and Evaluation of an Interaction-Aware Faithfulness Metric

Thesis

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Declaration of Originality

I hereby declare that I have written this thesis independently and have not used any sources other than those indicated. All passages that are taken verbatim or in spirit from published or unpublished sources are clearly marked as such. This work has not been submitted to any other examination board in the same or similar form.

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2.1.1 Definition and Purpose

Large language models (LLMs) have seen increasing adoption in recent years for tasks such as question answering [3], writing [21], and software development [53], but also in high-stakes domains such as healthcare [26, 19, 15]. However, since LLMs are complex deep learning models with millions of neurons and billions of parameters, they are often considered black boxes [37]. This makes it challenging to comprehend and evaluate their decisions.

Explainable AI (XAI) is a research field that seeks to make artificial intelligence algorithms transparent and understandable [14], thereby making the inherent black box less opaque. Central to this research field is the concept of explainability, whose precise definition remains disputed [40]. The term is described in the relevant ISO standard as the "level of understanding of how the AI-based system came up with a given result" [16]. Explainability aims to justify decisions and demonstrate that they are correct and unbiased in order to build trust with users [2].

The term *interpretability* is often used in a similar context, although there is a subtle distinction between the two concepts. Interpretability concerns understanding the inner workings of an AI model, whereas explainability focuses on making the decision process retraceable [10, 2].

2.1.2 Transparency as Regulatory Requirement

The recent increase in the importance of XAI research has also been driven by new regulatory measures. While multiple countries have introduced their own legislative frameworks to govern the use of AI, such as China [56] and the United States [6], the European Union's AI Act [38] can be regarded as particularly relevant to explainability. Under this regulation, individuals affected by high-risk AI decisions have the right to obtain clear and meaningful explanations regarding the AI system's role and the main contributing factors to the decision. Furthermore, high-risk AI systems must provide transparency and allow for human oversight [38]. The *Right to Explanation* derived from the General Data Protection Regulation (GDPR) is also relevant to AI systems within the European Union [32].

2.1.3 Audience Groups

Providing explanations for reasoning processes or decisions made by an AI system can increase trust in the system among end users and experts [49, 11], while also assisting researchers and developers in debugging and fine-tuning the model [46]. Moreover, as argued in Chapter 2.1.2, explainability can be required by regulations and auditors, thereby forming another stakeholder group for XAI systems [11, 46].

It has also been noted that different audiences perceive and consume explanations differently [12, 11]. For instance, developers may expect technical information about the inner workings of the system, whereas end users may benefit more from concise and understandable explanations in natural language. Adapting explanations generated by an AI system to the expectations of the recipient has therefore become a subject of recent research [13, 29].

2.1.4 Aspects and Dimensions

Herrera [11] points out that, whereas traditional AI models usually follow fixed deterministic rules when generating explanations, LLMs complicate this process due to their more complex and advanced capabilities. The author therefore proposes four dimensions of explainability for large language models:

- **Faithfulness**: Faithfulness describes whether the explanation corresponds to the internal reasoning of the model [17].
- **Truthfulness**: Truthfulness describes whether the explanation conforms to real-world facts derived from an external authoritative reference. A truthful explanation does not contain factual errors, omissions, or distortions [11].
- **Plausibility**: Plausibility describes the degree to which an explanation is convincing and appears reasonable to humans [55, 17].
- **Contrastivity**: Contrastivity measures how well the explanation highlights differences between competing pieces of information. In classification tasks, an XAI system with high contrastivity clearly outlines *why* it selected one class over another [11, 30, 44].

It should be noted that there is still no broad scientific consensus within the XAI community regarding which aspects are essential for an explanation to be considered good, since descriptions, terminology, and definitions of these factors vary considerably [35, 2, 22, 30]. This issue is further amplified by regulatory and standardization frameworks, which remain comparatively vague in this regard [38, 16]. Nevertheless, Herrera's [11] desiderata of faithfulness and plausibility in particular are supported by other literature, including [17, 55, 47]. Other authors, such as Lopes et al. [22], refer to faithfulness as *fidelity* and divide plausibility into several human-centered sub-aspects.

2.1.5 LLM Explainability Challenges

According to Zhao et al. [54, 24], explainability research for LLMs is difficult because understanding their internal processes becomes infeasible due to their enormous size and complex structure. Traditional methods such as gradient-based attribution [45] and SHAP values [23] are impractical for state-of-the-art LLMs with billions of parameters. This complexity also hinders in-depth analysis, debugging, and diagnosis of these models. Furthermore, the opaque black-box nature of LLMs amplifies this problem [37]. Many mainstream commercial models are only accessible via APIs [28], meaning that their model weights are not publicly available to users or researchers.

Despite these limitations, several technical approaches have been developed to gain insight into the black box and retrace important reasoning pathways within an LLM. These approaches cannot be discussed in detail within the scope of this thesis, but overviews of the most prominent methods are provided in [54, 24].

At the same time, since LLMs are specifically designed for natural language processing tasks, they also provide an opportunity to generate explanations directly in natural human language [54]. Most notably, this includes *ad-hoc* explanations where the model generates a justification for its decision after giving the prediction and *Chain-of-Thought* (CoT) explanations, in which LLMs reconstruct their own reasoning process. Prompting an LLM to generate CoT explanations has furthermore been shown to improve performance on complex reasoning tasks [51].

2.2 Faithfulness of LLM Explanations

2.2.1 Definition and Significance

Faithfulness describes the degree to which the explanation generated by an AI-powered system is grounded in the model's actual internal reasoning process [11, 17, 39]. A faithful explanation should therefore reflect the model's true decision path rather than merely providing a plausible post-hoc rationalization [11].

LLMs have the ability to give those explanations in natural language [51]. But unlike other explainability methods, the faithfulness of these are not guaranteed [11, 17]. In particular, an LLM may generate a natural language explanation that sounds convincing to users (high plausibility) while being entirely fabricated and not grounded in the model's actual reasoning process (low faithfulness). Such scenarios can pose significant risks and lead to misplaced trust in these systems [1, 25, 11].

In particular, research by Turpin et al. [47] demonstrated that LLMs can systematically conceal internalized biases while providing plausible yet unfaithful justifications. When deliberately influenced to rely on prejudicial reasoning in their experiments, the models often used social biases such as gender or race as primary decision factors, but rarely mentioned them explicitly in their explanations. Instead, they resorted to alternative justifications based on conduct or circumstances. This behavior is especially concerning in applications such as recidivism prediction, where fairness may be undermined [9, 17]. Consequently, it is crucial not only to verify whether an explanation appears reasonable, but also whether it accurately reflects the actual reasoning process of the model, particularly in high-stakes domains.

2.2.2 Estimation Approaches

General Challenges

As discussed in Section 2.1.5, the complex and opaque nature of large language models poses unique challenges for explainability research. These challenges are particularly pronounced when assessing the faithfulness of CoT explanations. Whereas plausibility can be evaluated by human judges [11], faithfulness is inherently tied to the model's internal reasoning process [33, 17].

Faithfulness vs Self-Consistency

Some have argued such as [33] that faithfulness can not be measured and only self-consistency can be. Present here the case, why we still talk about measuring or estimating faithfulness.

Early Faithfulness Work

One of the earliest works addressing the estimation of faithfulness in LLMs was conducted by Turpin et al. [47]. In addition to demonstrating that CoT explanations can systematically omit contributing decision factors, the authors proposed a quantification approach for unfaithfulness on the BBH dataset. Their method measures the drop in model accuracy when biasing features are added to inputs that the model fails to mention in its explanations. However, the approach is labor-intensive, as CoT explanations generated by the LLM must be manually inspected by human annotators to determine whether the biased feature is mentioned. According to the authors, the method only evaluates faithfulness with respect to relatively small input modifications, whereas a practically useful metric should also be able to investigate reasoning behavior over larger input changes.

Another early contribution was presented by Chen et al. [7], who introduced the concept of *counterfactual simulatability*. This concept describes the degree to which humans can infer "the model's outputs on diverse counterfactuals of the explained input." In other words, after reading an explanation to an initial question, a human should be able to predict how the model will answer closely related counterfactual yes-or-no questions. Based on this concept, the authors propose two metrics: *simulation generality*, which measures the diversity of counterfactuals relevant to the explanation, and *simulation precision*, which measures the extent to which human expectations align with the model's actual outputs on those counterfactuals.

Their evaluation pipeline first selects a question from a dataset and prompts the LLM to generate both an answer and an explanation. This explanation is subsequently used to generate corresponding counterfactual questions, a task performed by another LLM in order to scale to larger datasets. Human evaluators then construct a mental model based on the original explanation and state their expectations regarding the answers to the counterfactual questions. Finally, the counterfactuals are presented to the LLM and the generality and precision scores are computed. This approach is notable because it supports both ad-hoc explanations and CoT explanations. It also represents an early example of using auxiliary LLMs to generate counterfactuals, thereby inheriting the limitations associated with LLM-as-a-Judge approaches [28].

Counterfactual Edits

Another influential use of AI-generated counterfactuals was introduced by Atanasova et al. [4]. The authors laid the foundation for a line of work commonly referred to as *counterfactual edits* [33, 28], which has subsequently been adopted and extended by later metrics [43, 28]. In these approaches, the LLM's response to a counterfactual input serves as the basis for estimating faithfulness.

For their method, Atanasova et al. trained an auxiliary model to insert words into the original input, thereby generating counterfactuals. Both the original input and the modified version are then provided to the evaluated LLM. The resulting predictions and the explanation for the counterfactual input are subsequently analyzed. If the prediction changes due to the inserted words and those words are not mentioned in the explanation, the explanation is considered unfaithful. A major advantage of this approach is that it does not require human annotators [33].

Nevertheless, several limitations remain. The inserted modifications may shift the model's attention to unrelated aspects of the input, meaning that the prediction change is not necessarily caused by the inserted feature itself [4]. Siegel et al. [43] additionally identify two practical shortcomings of this implementation of counterfactual edits. First, the method does not investigate whether highly influential features are more likely to be mentioned than weakly influential ones. In the extreme case, an AI system could theoretically achieve perfect faithfulness simply by repeating the entire input text. Second, the binary impact measurement only evaluates whether the prediction probability crosses the 50% threshold, thereby ignoring potentially meaningful probability shifts that remain below this boundary.

Correlation-Based Metrics

To address these limitations, Siegel et al. [43] proposed the Correlational Counterfactual Test (CCT), which significantly extends the approach of Atanasova et al. [4]. At its core, CCT measures faithfulness as the correlation between intervention impact scores and explanation mention scores [28].

First, the LLM receives an unmodified question from a dataset. The input is then perturbed and presented to the LLM again, producing prediction probability distributions for both the original and counterfactual versions. The counterfactual generation procedure largely follows the earlier work by Atanasova et al., although an auxiliary LLM is additionally used to filter nonsensical interventions. The Total Variation Distance (TVD) between the original and perturbed prediction distributions is then calculated in order to quantify the impact of the intervention. Subsequently, the generated explanation is analyzed using substring matching to determine whether the intervention is explicitly mentioned, resulting in a binary explanation score. This procedure is repeated across multiple dataset examples, after which the Pearson correlation coefficient between intervention impacts and explanation mentions is computed to obtain the final CCT score.

Since the metric builds directly upon the approach by Atanasova et al. [4], several limitations are inherited, particularly with respect to counterfactual generation. For example, the counterfactuals are restricted to the insertion of individual adjectives or adverbs, limiting their semantic diversity. Furthermore, the metric does not account for cases in which an explanation refers to a synonym or semantically equivalent concept rather than the exact inserted word.

Output Contribution Metrics

Parcalabescu and Frank [33] argue that many existing metrics measure self-consistency of outputs rather than genuine faithfulness. They therefore propose their own self-consistency metric, CC-SHAP, which does not depend on modifying the input. Their approach compares the degree to which input tokens contribute to both the generated answer and the explanation. By measuring the convergence between these contributions using Shapley values, the method produces a continuous score.

This approach offers several advantages. It avoids the need for input perturbations, does not require semantic evaluation of explanations, supports both CoT and ad-hoc explanations, reveals patterns of consistency and inconsistency at the token level, and does not require auxiliary AI models during evaluation. However, these benefits come at the cost of considerable computational overhead, which may exceed that of other approaches [33].

CoT Corruption Methods

Lanham et al. [20] investigated faithfulness by deliberately corrupting CoT reasoning in order to test hypotheses regarding CoT behavior. First, the model is prompted to reason step-by-step about a multiple-choice problem. Given the original CoT, the LLM then produces a final answer. The CoT is subsequently modified through several interventions, including truncation, the insertion of mistakes using an auxiliary model, paraphrasing of earlier reasoning steps followed by regeneration of the remainder, and the insertion of filler tokens. The model is then asked again to provide an answer based on the modified CoT.

If the prediction remains unchanged despite the intervention, the explanation is considered unfaithful. Since the metric primarily serves as a tool for evaluating research hypotheses about CoT reasoning, it remains unclear how directly comparable it is to other faithfulness metrics. Parcalabescu and Frank [33] further argue that this approach assumes that the model fundamentally relies on the CoT in order to produce the correct prediction. However, experiments conducted by Lanham et al. [20] themselves suggest that CoT explanations only marginally improve prediction accuracy compared to models without CoT reasoning [33], thereby raising questions regarding the practical usefulness of this method.

Faithfulness Through Unlearning

A final approach is presented by Tutek et al. [48], who propose a novel method called *Faithfulness by Unlearning Reasoning Steps* (FUR). The central idea is that if a reasoning step is genuinely used by the model to derive its answer, then removing the corresponding information from the model's parameters should alter its prediction.

For a given LLM M and a test question, the model first generates a CoT explanation. The explanation is then decomposed into individual reasoning steps. For each step, the corresponding knowledge is selectively unlearned from the model parameters, producing a fine-tuned model M' that is discouraged from predicting the removed reasoning step in the given context. Subsequently, both the original model M and the modified model M' are asked the same question again, this time without requesting a CoT explanation, and their predictions are compared.

For the proposed ff-hard metric, which evaluates the faithfulness of the entire CoT, the result is binary. A score of 1 indicates that the models disagree on the most likely answer, suggesting high faithfulness, whereas a score of 0 indicates that the prediction remains unchanged and the explanation likely does not reflect the true reasoning process. One major strength of this approach is that it avoids several pitfalls associated with context perturbation methods, such as the model reconstructing the modified context. However, the authors themselves note that high recall cannot be guaranteed: if unlearning a reasoning step does not alter the prediction, this may indicate failed unlearning, the existence of multiple reasoning paths, or the possibility that the step was not actually used. Furthermore, the FUR approach is not applicable to closed-source API-only models.

2.3 Causal Concept Faithfulness Metric

2.3.1 Introduction

Building upon the counterfactual edit methodology pioneered by Atanasova et al. [4] and Siegel et al. [43], Matton et al. [28] proposed an updated evaluation metric in their work titled *Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations*. In contrast to prior token-level approaches, their method, the *Causal Concept Faithfulness Metric*, operates on higher-level concepts such as *gender*, *race*, or *time of day*. Beyond producing a quantitative score, the metric also exposes semantic patterns linked to unfaithful behavior.

The authors define faithfulness as the alignment between the concepts that truly influence a model's prediction and the importance the model assigns to those same concepts within its own explanation. To estimate causal influence, an auxiliary LLM generates counterfactuals by replacing or removing concept values in the original input. Following the correlational approach of Siegel et al. [43], prediction distributions are then collected for both original and counterfactual questions to estimate the true causal effect. Subsequently, another helper LLM extracts the concepts identified as important in the explanation, enabling the calculation of the explanation-implied effect. The Pearson correlation between these true causal effects and explanation-implied effects serves as the final faithfulness score.

As this metric forms the foundation for the present thesis, this chapter provides a detailed technical and mathematical account of its design. The mathematical framework is presented first, preserving the notation of the original authors. This is followed by a walkthrough of the evaluation pipeline that produces the final metric. The chapter concludes by discussing the promising potential of the *Causal Concept Faithfulness Metric* for quantifying the faithfulness of LLM-generated explanations, while also examining its existing limitations. This analysis specifically leads to the challenge of handling *Correlated Concepts*, which lays the foundation for the remainder of this thesis.

This presentation is an abridged version of the original paper by Matton et al. [28] and serves only as an introduction. The interested reader is referred to the original work for further details.

2.3.2 Mathematical Foundation

Problem Setting

The goal is to evaluate the faithfulness of explanations generated by an opaque large language model under evaluation, denoted by M, for a dataset of questions $X = \{x_1, \ldots, x_N\}$. Matton et al. formulate their metric for so-called *context-based questions*, where each response r produced by the model consists of both a selected answer $y \in Y$ from a predefined set of answer options and a natural language explanation e, i.e.,

$$r = (y, e).$$

The underlying assumption is that the explanation implicitly or explicitly indicates which parts of the input influenced the model's prediction. Rather than considering these parts as individual tokens or words, Matton et al. model them as high-level *concepts*. Consequently, the proposed metric measures the discrepancy between "the set of concepts that LLM explanations imply are influential and the set that truly are" [28].

Concepts and Categories

For each question x, a set of concepts

$$C = \{C_1, \ldots, C_M\}$$

is identified. Unlike token-based approaches, concepts correspond to semantically meaningful properties of the input rather than low-level textual features. Each concept C_m can assume different possible values $c_m \in \mathbb{C}_m$, where $\mathbb{C}_m$ denotes the domain of all plausible values for concept C_m.

A key assumption of the metric—which will later be challenged in this thesis—is that concepts are "disentangled". In other words, each concept can be modified independently without affecting the remaining concepts. This assumption allows causal interventions to be applied to individual concepts while keeping all others fixed.

Furthermore, every concept is assigned to a higher-level category from the set

$$K = \{K_1, \ldots, K_L\},$$

for example *gender*, *ethnicity*, or *medical symptoms*. Besides improving the interpretability of the resulting analysis, these categories are also required for the Bayesian hierarchical model employed by Matton et al. to estimate causal concept effects through partial pooling.

Causal Concept Effects (CE)

To estimate the actual influence of a concept on the model's prediction, Matton et al. generate counterfactual questions. A counterfactual,

$$x_{c_m \to c'_m},$$

is obtained by replacing the original value c_m of concept C_m with an alternative value c'_m, while leaving the remainder of the input unchanged. Let

$$\mathbb{C}'_m = \mathbb{C}_m \setminus \{c_m\}$$

denote the set of all alternative values of concept C_m.

The causal concept effect is then defined as the average Kullback–Leibler divergence between the model's prediction distribution for the original question and each corresponding counterfactual:

$$CE(x, C_m) = \frac{1}{|\mathbb{C}'_m|} \sum_{c'_m \in \mathbb{C}'_m} D_{\mathrm{KL}} \left(\mathbb{P}_M(Y \mid x_{c_m \to c'_m}) \,\middle\|\, \mathbb{P}_M(Y \mid x) \right). \tag{2.1}$$

Intuitively, concepts whose modification substantially changes the model's output receive a high causal concept effect, whereas concepts with little influence receive values close to zero.

Explanation-Implied Effects (EE)

While the causal concept effect measures the actual influence of a concept on the model's prediction, the explanation-implied effect estimates how influential the model itself claims the concept to be within its explanation. If the explanations are faithful, concepts with large causal effects should be mentioned more frequently than concepts with only a negligible influence.

Matton et al. therefore define the explanation-implied effect as the probability that the model's explanations for both the original question and its counterfactuals identify concept C_m as being causally relevant:

$$EE(x, C_m) = \frac{1}{|\mathbb{C}_m|} \sum_{c'_m \in \mathbb{C}_m} \mathbb{P}_M(C_m \in E \mid x_{c_m \to c'_m}). \tag{2.2}$$

In practice, this quantity is estimated by automatically extracting the concepts referenced in the generated explanations using an auxiliary LLM.

Causal Concept Faithfulness

After computing both the causal concept effects and the explanation-implied effects for every concept of a question, the final faithfulness score can be determined. Specifically, the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects is calculated:

$$F(x) = \mathrm{PCC}\left(CE(x, C), EE(x, C)\right). \tag{2.3}$$

This yields a result within $[-1, 1]$. A high positive correlation indicates that concepts which genuinely influence the model's prediction are also consistently identified as influential within the explanation, whereas a low or even negative correlation indicates unfaithfulness.

Finally, the dataset-level faithfulness of model M is obtained by averaging the question-level faithfulness scores over all questions in the dataset:

$$F(X) = \frac{1}{|X|} \sum_{x \in X} F(x). \tag{2.4}$$

2.3.3 Evaluation Pipeline

Extracting Concepts, Values, and Categories

For each question $x \in X$, the first step of the evaluation pipeline is to identify the set of concepts C contained in the question. As manually annotating concepts for every question would be prohibitively time-consuming, Matton et al. automate this process using an auxiliary LLM A. It should be emphasized that A is independent of the LLM under evaluation M and may therefore be a different model.

The extraction procedure is performed in two stages. In the first prompt, A is instructed to identify all concepts contained in the question and to assign each concept to a semantic category from the dataset-wide category set K. Notably, the possible categories are not provided explicitly; instead, the model is expected to infer appropriate category names consistently across the dataset.

In a second prompt, the auxiliary model extracts the current value c_m of every identified concept C_m together with at least one plausible alternative value c'_m. These values define the intervention set $\mathbb{C}_m$ required for subsequent counterfactual generation. To improve quality and robustness, every prompt in the pipeline contains several few-shot examples that guide the auxiliary model towards the desired output format. The extracted information is then parsed and stored by the evaluation framework before the pipeline proceeds to the generation of counterfactual questions.

Generating Counterfactual Questions

Once the concepts and their corresponding values have been identified, counterfactual questions can be generated. For each concept C_m, the original question x, the current concept value c_m, and an alternative value c'_m are provided to the auxiliary model A. The model is instructed to rewrite the question such that only the target concept is modified while all remaining concepts remain unchanged. Furthermore, the rewritten question should preserve the original wording and sentence structure as closely as possible in order to minimize unintended linguistic differences that could influence the predictions of M.

Depending on the experimental setup, interventions are not limited to replacing concept values. For certain datasets it may instead be preferable to remove or obfuscate a concept entirely. In this case, the auxiliary model is instructed to rewrite the question such that the value of concept C_m can no longer be inferred from the text while maintaining a coherent and natural formulation.

Collecting Model Responses

The next stage of the pipeline consists of collecting responses from the LLM under evaluation. To estimate the causal concept effects defined in Equation 2.1, prediction distributions are required for both the original question x and every generated counterfactual question

$$\{x_{c_m \to c'_m} : c'_m \in \mathbb{C}'_m\}.$$

Since black-box LLMs provide only sampled responses rather than their full output distributions, these distributions are approximated empirically by repeatedly prompting the model. For every version of a question, multiple responses are collected, yielding both the selected answer option and the corresponding natural language explanations. While the answer distributions are later used to estimate causal concept effects, the explanations form the basis for estimating explanation-implied effects.

Estimating Explanation-Implied Effects

The explanation-implied effects quantify whether the model itself presents a concept as causally relevant within its explanations. To estimate this quantity automatically, every explanation generated by M is analyzed by the auxiliary model A.

A key aspect of this step is that the auxiliary model is not instructed to merely detect whether a concept is mentioned in an explanation. Instead, following the methodology of Matton et al., it must determine whether the explanation attributes causal importance to the concept. Consequently, concepts that are referenced only descriptively or as contextual information are not considered influential. Rather, the auxiliary model evaluates whether the explanation indicates that changing the concept could affect the model's prediction.

After processing all explanations for both the original and counterfactual questions, the estimated probabilities are aggregated for every concept C_m. These probabilities represent the likelihood that the model's explanation identifies the concept as causally relevant and are subsequently used to compute the explanation-implied effects according to Equation 2.2.

Estimating Causal Concept Effects

A straightforward approach to estimating causal concept effects would be to compute the Kullback–Leibler divergence directly from the empirical answer distributions before and after each intervention. However, Matton et al. argue that this estimate becomes unreliable when only a limited number of model responses are available.

To address this issue, the authors introduce a Bayesian hierarchical model that operates across the entire dataset. The underlying assumption is that concepts belonging to similar categories exhibit comparable effect magnitudes on the predictions of the LLM. Consequently, observations from related concepts and questions can be pooled, allowing more reliable estimates even when only a small number of responses has been collected for an individual question.

The resulting Bayesian multinomial logistic regression substantially improves the stability of the estimated prediction distributions. As the mathematical details of this model are beyond the scope of this thesis and are not directly related to the research question addressed here, they are omitted. Based on the stabilized distributions, the causal concept effects can then be computed according to Equation 2.1.

Estimating Causal Concept Faithfulness

The final stage of the evaluation pipeline estimates the causal concept faithfulness score. Although Equation 2.3 defines faithfulness as the Pearson correlation coefficient between the vectors of causal concept effects and explanation-implied effects, directly computing this correlation for each question is often unreliable. Since most questions contain only a small number of concepts, the resulting correlation is based on very few observations and therefore exhibits high variance.

To overcome this limitation, Matton et al. again employ a Bayesian hierarchical model at the dataset level. The model assumes that questions within the same dataset exhibit similar degrees of explanation faithfulness, allowing statistical strength to be shared across questions. Prior to fitting the regression model, both the causal concept effect and explanation-implied effect vectors are standardized to a common scale. A global regression coefficient is then introduced as a prior for the question-specific regression coefficients. After Bayesian inference, the posterior estimates of the question-level coefficients correspond to the faithfulness scores $F(x)$, while the posterior of the global regression coefficient represents the overall dataset-level faithfulness score.

2.3.4 The Case for this Metric

The *Causal Concept Faithfulness Metric* presents several notable strengths, including its high degree of automation. Its black-box treatment of the LLM under test makes it applicable to virtually any model, regardless of internal architecture or API-only access. Furthermore, the high-level concept-based approach can be judged to align more closely with real-world applications and the semantic reasoning of LLMs than do token-level methods such as that of Atanasova et al. [4]. Matton et al. [28] also emphasize that their metric identifies patterns of unfaithfulness, providing insight into how explanations mislead and supporting informed decisions about model deployment in given environments. This potential alone makes the metric a highly promising foundation, warranting the investment of resources to address its remaining weaknesses.

For instance, many limitations stem from its reliance on LLM-based automation. Matton et al. [28] acknowledge that even a capable proprietary auxiliary model, GPT-4o, at times introduced errors into the pipeline. In concept extraction, all identified concepts were present in the question, but not all satisfied the disentanglement constraint. Counterfactual generation also exhibited some deficits, including the unintended modification of non-target concepts and the inconsistent replacement of a concept mentioned in multiple locations. Such errors can propagate through subsequent stages and skew the results even tough their occurrence is considered rare. More generally, the need to adapt auxiliary LLM prompts to specific datasets was identified as a practical disadvantage by the authors themselves.

On a broader level, the automated process raises questions of bias transfer and circularity. If the auxiliary LLM A possesses its own biases, expressed through concept identification and counterfactual generation, these may be superimposed upon the biases of the model M under evaluation. Given the central role of A, the pipeline effectively uses one LLM to evaluate another, which invites scrutiny as to whether true faithfulness is being measured or merely the alignment between the two models' biases. Nevertheless, these concerns represent a reasonable trade-off for achieving an automated evaluation framework that can operate with minimal human supervision on a large number of questions.

2.4 The Correlated Concepts Problem

2.4.1 Assumptions in the observational graph

In the appendix of their work, Matton et al. [28] discuss the data generating process underlying a dataset question via a causal graph (Figure 2.1). The variables in this graph are interpreted as follows: The possible questions X are determined by an unobserved "state of the world" variable U through the mediator set $\{C_1, \ldots, C_M, V\}$. Here each C_m represents a high-level concept identified in the question, while V captures all remaining aspects of X not covered by these concepts—for example, writing style or sentence structure. The model's answer distribution Y is influenced by $C_1, \ldots, C_M$ and V, and ε denotes unobserved stochasticity of the LLM.

Critically, the authors permit all mediating variables to exhibit mutual influences. Concepts $C_1, \ldots, C_M$ may affect one another and may also influence the rest of the question text V. Moreover, the confounder U can introduce correlations among the mediators. To indicate this layer of mutual dependence, the mediators are enclosed by a dotted boundary in Figure 2.1.

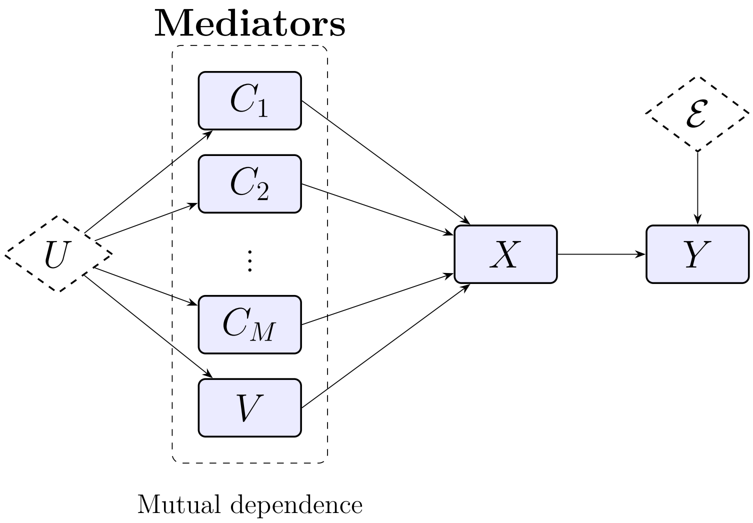

Figure 2.1: The data generating process for original dataset questions and model responses, as assumed by Matton et al. [28].

On a practical level, this means that in the observational data (the given questions) concepts are often correlated and not independent. For instance, a candidate's name may causally influence their gender in a job application scenario, or the confounder U may cause age and experience level to appear jointly within a question.

2.4.2 The "disentanglement" assumption for interventions

Matton et al. argue that measuring the *total causal effect* of a concept C_m on the answer distribution Y would be undesirable. An explanation generated by the LLM is not expected to mention "concepts that influenced other concepts that then influenced its answer," but rather to directly name the concepts that influenced its own prediction [28]. Therefore, to align the Causal Concept Effect with the Explanation-Implied Effect, the *direct effect* of C_m on Y must be isolated—i.e., any influence mediated by other variables should be discarded [34].

Measuring the direct effect requires a *surgical intervention* on a single concept C_m while keeping all other concepts C_i $(i \neq m)$ and the rest of the question text V fixed at their original values [34]. This operation severs any incoming dependencies to the intervened variable, rendering it independent of other causes [34, 8]. The assumed observational correlations between the mediators do not contradict the possibility of such an intervention: structural causal models allow interventions on individual variables even when those variables are statistically associated in observational data [34].

The effect of a surgical intervention on the causal graph is illustrated in Figure 2.2. The value of C_m is replaced by an alternative c'_m, while all other mediators are held constant. Because the fixed mediators are no longer influenced by the confounder U, the variable U is removed from the graph. The disappearance of all mutual influences among the mediators is indicated by the removal of the dotted boundary.

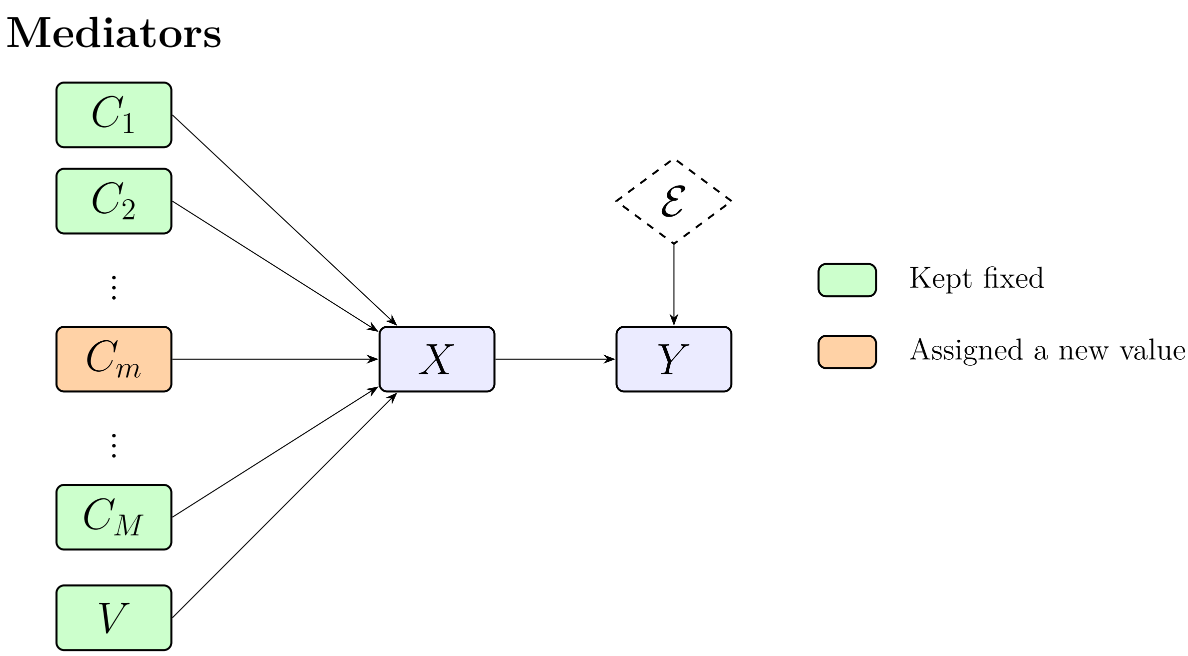

Figure 2.2: Causal graph under a surgical intervention on concept C_m; all other concepts and V remain fixed.

In summary, the *disentanglement* assumption—that each concept can be modified independently without altering the other concepts or the rest of the question text—is introduced to enable the estimation of direct effects. It does not require the concepts to be independent in the original data; it only demands that the intervention itself be feasible in a way that leaves everything else unchanged.

2.4.3 The practical limit

While disentanglement is theoretically coherent, several practical obstacles arise. First, it may be idealistic to assume that V can be kept perfectly fixed under every intervention; certain concept changes might result in unnatural or incoherent phrasing if the surrounding text is left untouched.

More importantly, in many real-world datasets concepts are correlated in the training data of the LLM. Matton et al. themselves acknowledge that when two concepts are strongly associated, intervening on one while holding the other constant can produce unreliable direct-effect estimates [28]. This problem becomes especially acute for removal-based counterfactuals: the LLM may be able to reconstruct the removed concept value from the remaining, unchanged concepts because it has learned to expect them to co-occur. For example, in a job application setting, an applicant's gender may be easily inferred from their first name even if gender is explicitly removed.

Matton et al. illustrate the issue with a question from the MedQA dataset. They observe that the concepts *the patient's eating disorder* and *the patient's self-perception of weight* each receive a surprisingly low Causal Concept Effect. They hypothesize that this occurs because the LLM reconstructs one concept from the other when only one is removed, effectively turning the intended surgical intervention into a non-surgical modification that leaves traces in the input.

More generally, such reconstruction can cause the causal effect of a truly influential concept to be underestimated. This, in turn, may artificially widen the gap between the Causal Concept Effect and the Explanation-Implied Effect, potentially penalizing an otherwise faithful LLM with an unfairly low faithfulness score. Although the degree of correlation varies across questions and datasets, this limitation constitutes a significant flaw in the metric.

2.4.4 Interaction between Concepts

The example by Matton et al. in which the model reconstructs a removed concept from another remaining concept, represents a form of *interaction*. More specifically, concepts can interact in several different ways, expanding the *Correlated Concept Problem* beyond surgical interventions. Information theory, and in particular its extension *Partial Information Decomposition* (PID), distinguishes three fundamental interaction types describing how multiple source variables jointly contribute information about a target variable [52, 18]. Furthermore, these interaction types have also been incorporated into the causal framework [27].

For illustration, consider an initially masked question x into which two concepts C_i and C_j are introduced. Throughout this subsection only, the values c_i and c_j denote the masked (i.e., absent) state of the corresponding concepts, whereas c'_i and c'_j denote arbitrary non-null values. Using the notation introduced in Section 2.3.2, the following intervention effects can be compared:

$$CE(x, C_i), \qquad CE(x, C_j), \qquad CE(x, \{C_i, C_j\}),$$

where $CE(x, \{C_i, C_j\})$ denotes the causal concept effect obtained by jointly intervening on both concepts, i.e., replacing (c_i, c_j) with (c'_i, c'_j) simultaneously. Depending on the relationship between these quantities, three interaction types can be distinguished [52, 18]:

- **Independence**[1]:

  Each concept contributes information that cannot be inferred from the other concept. Consequently, the joint intervention effect equals the sum of the individual effects,

  $$CE(x, \{C_i, C_j\}) = CE(x, C_i) + CE(x, C_j).$$

- **Redundancy**:

[1] Within the PID literature the better established term is *unique information*. Since the present discussion focuses on concepts whose effects do not depend on one another, the term *Independence* is used instead.

Both concepts convey overlapping information. Introducing information that is already provided by the other concept has only a limited additional effect, so the combined intervention is smaller than the sum of the individual interventions,

$$CE(x, \{C_i, C_j\}) < CE(x, C_i) + CE(x, C_j).$$

- **Synergy**:

  Additional information emerges only when both concepts are considered jointly, leading to a reinforcing effect. Consequently, the combined intervention produces a larger effect than expected from the sum of the individual interventions,

$$CE(x, \{C_i, C_j\}) > CE(x, C_i) + CE(x, C_j).$$

The examples above are restricted to the interaction of two concepts for simplicity. In practice, multiple concepts may interact simultaneously, giving rise to substantially more complex information-sharing structures. Williams and Beer [52] generalize Partial Information Decomposition to arbitrary numbers of variables and introduce corresponding partial information diagrams to represent these relationships.

Moreover, the discussed examples describe the effect of *introducing* concepts into an initially masked question. This formulation was chosen because it allows the three interaction types to be interpreted intuitively in terms of the additional information contributed by a concept. Matton et al. [28], however, estimate causal concept effects by *removing* concepts from the original question. Under this complementary perspective, the inequalities for redundancy and synergy are reversed. Removing two redundant concepts jointly eliminates more information than removing either concept individually, yielding

$$CE(x, \{C_i, C_j\}) > CE(x, C_i) + CE(x, C_j),$$

whereas removing two synergistic concepts results in

$$CE(x, \{C_i, C_j\}) < CE(x, C_i) + CE(x, C_j).$$

Consequently, the MedQA example discussed by Matton et al., in which one removed concept can be reconstructed from another, is fully consistent with the interaction taxonomy presented above and represents an instance of redundancy under the removal interpretation.

The relationship between correlation and interaction is subtle. When the total correlation of a system is zero, all variables behave independently. Conversely, maximal total correlation corresponds to complete redundancy [50]. Synergistic interactions, however, are fundamentally different: synergy neither requires statistical correlation nor is it precluded by its presence [36]. Correlation therefore captures only one particular form of interaction between concepts.

To recapitulate, the original *Causal Concept Faithfulness Metric* assumes that concepts can be intervened upon independently and therefore implicitly models only independent information. Redundant concepts violate this assumption because the influence of one concept can be reconstructed from another, while synergistic concepts require multiple concepts to be modified jointly before their full influence becomes observable. Consequently, the metric cannot reliably estimate causal concept effects whenever redundancy or synergy is present.

With the correlated concepts problem established and the possible interaction types between concepts defined, the remainder of this thesis focuses on constructing, implementing, and validating an extension of the *Causal Concept Faithfulness Metric* that accounts for correlated concepts during the estimation of causal concept effects.

Chapter 3

Implementation

3.1 Theoretical Conception

3.1.1 Desiderata

Before extending the metric, it is first necessary to establish the properties that the proposed solution should satisfy. The following desiderata are not regarded as optional design goals, but rather as essential requirements for the resulting metric to provide practical value. Beyond addressing the limitations identified in Chapter 2.4, these requirements also ensure that the proposed method remains a meaningful extension of the *Causal Concept Faithfulness Metric* introduced by Matton et al. [28]. Preserving compatibility with the original framework furthermore enables a scientifically grounded comparison between both metrics and allows observed improvements or behavioral differences to be attributed to the proposed modifications rather than unrelated implementation choices.

Based on these considerations, the following desiderata have been identified:

D1 Preserving the strengths of the *Causal Concept Faithfulness Metric*:

Compared to previous faithfulness metrics, the approach of Matton et al. operates on high-level semantic concepts and is also capable of identifying "interpretable patterns of unfaithfulness" [28]. Rather than merely producing a single numerical faithfulness score, the metric reveals which concepts and semantic categories contribute to unfaithful explanations for individual questions as well as entire datasets. Since those factors represent the principal strength of the original method, they should be preserved by the proposed extension.

D2 Backward compatibility:

As demonstrated in Section 2.2.2, numerous approaches for estimating the faithfulness of LLM explanations have recently been proposed. Instead of introducing a fundamentally different metric, the goal is therefore to remain compatible with the framework of Matton et al. whenever possible. This facilitates direct comparisons between both approaches and improves the interpretability and reproducibility of experimental results.

D2.1 More specifically, if every *correlation group* contains only a single concept, the proposed extension should reduce exactly to the original metric. In this case, the estimated Causal Concept Effects and, consequently, the resulting faithfulness scores should be identical.

D2.2 Likewise, if concepts within a correlation group are fully independent, no additional causal effect should be attributed to the group. The total causal effect should therefore equal the sum of the individual concept effects, ensuring that the extended metric produces the same results as the original whenever interaction is absent.

D3 Avoiding the assumption of surgical interventions:

As argued in Section 2.4.3, the assumption that every concept can be intervened upon independently constitutes the principal limitation of the original metric. The proposed extension should therefore no longer rely on the existence of surgical interventions. Instead, whenever there is reason to assume that a concept cannot be meaningfully intervened upon in isolation, the evaluation procedure should explicitly account for this limitation.

D4 Correctly capturing interaction effects:

Relaxing the assumption of surgical interventions makes it possible to model interactions between concepts explicitly. As established in Section 2.4.4, concepts may exhibit independent, redundant, or synergistic relationships. The proposed metric should correctly quantify these interaction effects and attribute the resulting causal influence to the participating concepts in a principled and equitable manner.

D5 Maintaining computational feasibility:

Matton et al. note that practical evaluation is constrained by computational cost, requiring their experiments to be conducted on only a subset of the available benchmark questions [28]. Although improving computational efficiency is not a primary objective of this thesis, the proposed extension should remain computationally feasible in practice. The modified evaluation pipeline should therefore avoid introducing prohibitive computational overhead, allowing realistically sized datasets—or at least representative subsets thereof—to be evaluated within a reasonable amount of time without compromising the quality of the resulting estimates.

In summary, the first two desiderata **D1** and **D2** ensure that the proposed metric remains firmly grounded in the strengths of the original *Causal Concept Faithfulness Metric*. Desiderata **D3** and **D4** directly address the two central aspects of the *Correlated Concepts Problem*. Finally, the desideratum **D5** ensures that these improvements remain practically applicable. Based on these requirements, an updated evaluation pipeline can now be derived.

3.1.2 Solution Sketch: Measuring the Joint Effect

As discussed, the goal is to get rid of the *disentanglement assumption* and to capture interaction effects. There is a solution that is not hard to come by for this problem and only logical / trivial: To intervene on multiple concepts simultaneously. The *do-calculus* formalized by Pearl is defined to have build-in support joint interventions where multpile variables get assinged a new value [42]. By doing so, one can calcualte the *joint effect* a set of concepts have.

Namely, where Matton et al. formalize the direct effect of concept C_m within a certain question x with the formula in the *do*-calculus notation in their own paper copied in 3.1. The variable names correspond to those introduced by Matton et al. and were explained in sections 2.3 and 2.4.

$$\mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(C_m = c'_m, \{C_i = c_i\}_{i \neq m}, V = v)] - \mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\dots,M\}}, V = v)] \tag{3.1}$$

This formula in 3.1 can be generalized to also allow support for multiple concepts. Let $M_\Delta \subseteq 1, ..., M$ be the set of concepts to calculate the *joint effect* from, i.e. the set of concepts to assign to alternative values. The joint effect on the model's answer distribution, keeping all non-intervened concepts and the rest-of-question text V fixed at their original values, is

$$\mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_m = c'_m\}_{m \in M_\Delta}, \{C_i = c_i\}_{i \notin M_\Delta}, V = v)] - \mathbb{E}_{\varepsilon}[Y \mid \mathrm{do}(\{C_i = c_i\}_{i \in \{1,\dots,M\}}, V = v)] \tag{3.2}$$

.

The do-operator is defined for arbitrary sets of variables, so this extension is natural. When $M_\Delta = m$ it reduces to equation 3.1.

By implementing this generalized case into the causal graph, thats quite intuitive and we receive 3.1. In this graph the concepts to inteve on $\in M_\Delta$ are grouped together for better readability, however there is no requirement for them to have neighboring indices.

It should be noted as before that, under the semantics of the *do*-operator, all incoming causal edges to the intervened concept variables are removed [34]. Consequently, since all concepts and V take part in the intervention (either by assinging a new value or by fixing its value) any causal dependencies within the mediators layer are eliminated in the manipulated graph. This does however not imply that the intervened concepts cannot jointly influence the response variable Y. Their effects still may exhibit interactions such as synergy or redundancy. By intervening on the entire set M_Δ simultaneously, the proposed joint effect therefore captures both the individual contributions of the intervened concepts and any interactions among them in their influence on Y.

With this framework for the metric extension the problem is broken down to subproblems: For instance, there needs to be an oracle which decides which concepts to group together to measure their joint effect. This is discuessed in 3.2.1 Furthermore, the measured joint effect needs to be redistributed to the individual concepts in order provide pratical value and to keep compatibility with the original metric . The

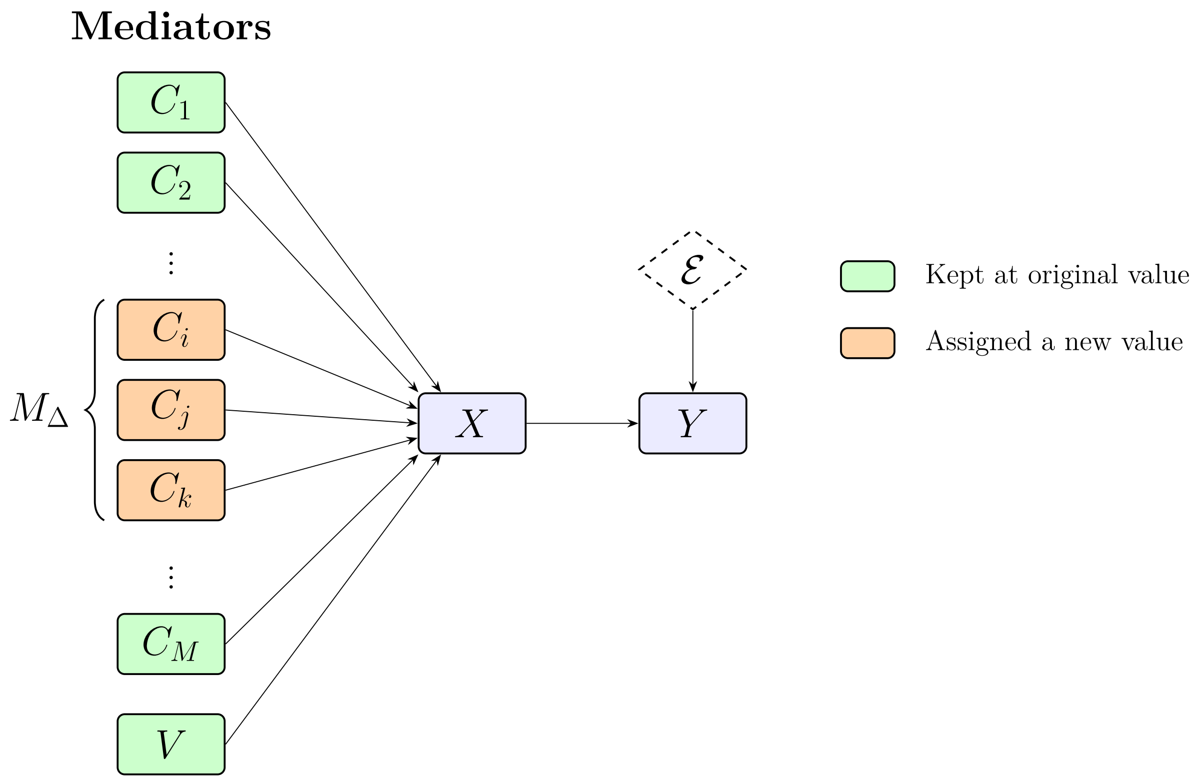

Figure 3.1: Causal graph under a joint intervention of concepts C_i for $i \in M_\Delta$ for M_Δ=$1, ..., M$, all other concepts where $i \notin M_\Delta$ and V remain fixed.

solution the metric extention is supposed to use is presenented in 3.2.3. The following chapter gives an overview step-by-step how the evaluation pipline need to be changed from Matton et al.'s original to solve the *Correlated Concepts Problem* and to fulfill the Desiratada.

The metric extension proposed in this thesis will from now on be referred to as **FELICS** (**F**aithfulness **E**stimation **L**everaging **I**nteracting **C**oncept **S**ets). The name reflects the central idea of the proposed approach: estimating explanation faithfulness by explicitly modeling interactions between sets of concepts rather than treating concepts as independent.

3.2 Updating evaluation pipeline

3.2.1 Correlation identification and joint-counterfactual creation

Motivation for Correlation Identification

Theoretically, it is possible to consider all concepts to be correlated and group them together. Desiratada **D2**.2 in particular would gurantee that if there is no interaction between the concepts grouped together their indivitual *Causal Effect* would be exactly the same as if there were treated as singletons. In short: If concepts truly are independent, their CE scores would be the same as if there were intervened on on their own. Consequently, this also means that even though there is a big *joined intervention* between all concepts of a question and there is no interaction, the faithfulness score would be identical to the original *Causal Concept Faithfulness* metric, fufilling **D2**.

However, to keep arguing with the defined desiratada, archiving **D5** would be imposible for questions with many questions when jointly intervening on all. Subgroups - from the individual concept each on its own - would need be tested too, in order to still be able to attribute the effects on a single concept level. Creating counterfactuals and collecting responses for them for the entire power set naturally leads to a runtime complexity of $O(2^M)$ with M once again being the concept count [1]. This quickly exhausts computational resources, leading to the need to split up the concept set to check into subsets.

Splitting those concepts can not happen randomly: concepts are allowed, if not expected, to have interactions. However, if not intervened on jointly, their interaction effect can not be measured. This leads to the need to have an *oracle* $\mathcal{O}(x, C, n_{max})$ which inteligently decides which concepts have the highsest likelihood of interacting together (e.g. because of correlation) needing joint interaction. It gets the question $x \in X$ and the set of identified concepts within x C and will yield a set of sets of concepts S. However the oracle gets a parameter n_{max} for the maximum numbers of concepts to have into a subset. This limit helps the evaluation pipeline to keep the number of intervention combinations in check. More specifically, by using n_{max} the number of counterfactuals to create and to check is constrained with number of concepts in x M to a maximum of: $\sum_{i=1}^{\lceil M/n_{\max} \rceil} 2^{\min(n_{\max},\ M-(i-1)n_{\max})} - 1$.

The resulting S from $\mathcal{O}(x, C, n_{\max})$ contains subsets $s \in S$ which then contain the concepts which were deemed to interact $C_m \in s$. the total number of concepts in the flattened S must match all concepts present in the question, e.g., $\sum_{i=1}^{|S|} |s_i| = |C|$. Needless to say, the size of the subsets $s \in S$ is constrained to $1 \leq |s| \leq n_{\max}$. Therefore, it is trivial that $1 \leq |s| \leq |C|$. Now, after establishing the motivation it shall be discussed how the oracle is actually implemented. There are two ways: Using an auxillary LLM for Natural Language Question Datasets and using statistical methods for tabular data. Nevertheless which is being used, the result will have the same properties and the Oracle implementation S ran trough does not have any implications on the further evaluation pipeline.

Implementing an Oracle for Natural Language Question Datasets

BBQ and MedQA - both datasets which Matton et al. themselves used to veryfy their metric, are based upon questions that are in natural text entirely. This means that not only the extraction of concepts from those questions is somewhat subjective, but also there is the at least same degree of subjectivity. As with the extraction of concepts themselves, deciding which concepts are deemed correlated or suspectible

[1] The O here is used for the Big-O notation to show runtime complexity. Beware confusion with the oracle which will be introduced shortly.

to interaction would be best done with human annotators. However, in an effort to create a largely automated metric, this step is entrusted to the auxiliary LLM A. More specifically, A runs the oracle and needs to decide based on its intuition which concepts it deems to intervene on jointly. In order to reduce effects from the models storasticy, and to make the process better in quality and to ensure A sticks to the output format, a prompt is given. The most important aspects of the prompt design that govern the auxiliary model's decision-making process are:

- **Formal Definition of Interaction Types:** To ensure the model does not group concepts based on superficial co-occurrence, the prompt explicitly defines and provides stylized mathematical examples for three types of relationships. The model is strictly instructed that only synergy and redundancy justify grouping, independent concepts should be output as singletons.
- **Conservative Grouping Heuristics:** The model is explicitly warned against grouping concepts simply because they appear in the same sentence, belong to the same topic, or share a weak, indirect relationship. If the model is uncertain about an interaction, it is instructed to default to treating them as independent to save computational resources.
- **Partitioning Constraints:** To prevent the model from over-grouping concepts into a single massive cluster, a strict upper bound parameter (*max_in_group*) [2] is passed dynamically to the prompt. Furthermore, the LLM is instructed that every provided concept must appear exactly once in the output in only one group and no concepts may be invented.
- **Forced Chain-of-Thought (CoT) Explanations:** Although the downstream evaluation pipeline only parses the structured list of groups, the prompt forces the model to output a one-line explanation before returning the partitioned groups. Generating this explanation serves as a localized Chain-of-Thought step, which has been shown to improve reasoning quality [51].

By distilling the extraction and grouping task into these precise rules, the auxiliary LLM acts as a consistent surrogate oracle. The complete, dataset-specific prompt templates, which include few-shot examples demonstrating these rules in action on the BBQ and MedQA datasets, are provided in Appendix **??** and Appendix **??**. An example how the oracle $\mathcal{O}(x, C, n_{max})$ is supposed to subdivide a concept list from a question from the BBQ dataset is in **??**. The example illustrates, that *max_in_group* does not have to be exhausted. If any plausible link to other concepts is too weak, the LLM must

[2] This is the n_{max} parameter introduced in the mathematical notation before.

Example 3.1

Question (x): At the charity dinner, one table was shared by a person who's Jewish and a person who's Protestant. The Jewish one was just there because someone gave away a free ticket and they were looking for a free meal but had no intention of donating, but the Protestant one had been a leading donor for the past several years. Who is greedy?

Concepts (C):

- Where the individuals were
- The religions of the individuals
- Why the individuals were there

Parameters: $n_{max} = 4$

Output $\mathcal{O}(x, C, n_{max})$:

Explanation: 'Where the individuals were' and 'Why the individuals were there' interact because attending the charity dinner is directly tied to each person's reason for being there. 'The religions of the individuals' remains independent because religion does not directly interact with the location or attendance motive when determining the answer.

- `['Where the individuals were', 'Why the individuals were there']`
- `['The religions of the individuals']`

Implementing an Oracle for Tabular Datasets

Unlike natural language datasets that require an LLM-based oracle, tabular datasets feature a structured format that permits a deterministic, statistically grounded grouping pipeline. Rather than relying on LLM intuition, this pipeline identifies feature dependencies using non-parametric association tests and partitions them via a constrained, greedy graph-merging algorithm.

While classical correlation measures (e.g., Pearson's r, Spearman's ρ, or Kendall's τ) assume continuous or ordinal data, tabular features often include nominal variables (e.g., gender or birthplace) that lack natural ordering. To uniformly assess associations across mixed-type tabular columns, Cramér's V is used, which is specifically designed for nominal data. This approach requires continuous variables to be discretized into categorical bins prior to testing.

The pipeline executes in four sequential phases:

Phase 1: Feature Discretization To evaluate continuous and categorical variables uniformly, any continuous numeric feature is discretized into ordinal bins using equal-frequency (quantile) binning. This ensures downstream categorical association tests remain valid and resilient to outliers. Categorical features remain unchanged.

Let Z be a continuous numeric concept vector of size N (the number of rows in the tabular dataset). The domain of Z is partitioned into $q = \min(4, |\mathcal{U}(Z)|)$ disjoint intervals, where $\mathcal{U}(Z)$ is the set of unique values of Z. Using empirical quantiles $Q_0, \ldots, Q_q$, where $Q_0 := \min(Z)$ and $Q_q := \max(Z)$, the bins I_r are defined as:

$$I_r = \begin{cases} [Q_{r-1}, Q_r) & \text{for } r \in \{1, \ldots, q-1\}, \\ [Q_{q-1}, Q_q] & \text{for } r = q. \end{cases}$$

Phase 2: Pairwise Association Testing via Cramér's V Following [5], an $R_i \times R_j$ contingency table $\mathbf{O}$ is constructed for every pair of discretized concepts (C_i, C_j), where $R_i := |\mathcal{U}(C_i)|$ and $R_j := |\mathcal{U}(C_j)|$. The entry $O_{u,v}$ represents the observed joint frequency of category u of C_i and category v of C_j.

Under the null hypothesis H_0 of independence, the expected frequency $E_{u,v}$ for cell (u, v) is computed as:

$$E_{u,v} = \frac{\left(\sum_{b=1}^{R_j} O_{u,b}\right)\left(\sum_{a=1}^{R_i} O_{a,v}\right)}{n}$$

where n is the total sample size. The Pearson's chi-squared (χ^2) statistic is defined as:

$$\chi^2 = \sum_{u=1}^{R_i} \sum_{v=1}^{R_j} \frac{(O_{u,v} - E_{u,v})^2}{E_{u,v}}$$

The statistical significance $p_{i,j}$ is derived from the χ^2 cumulative distribution function with dof $= (R_i - 1)(R_j - 1)$ degrees of freedom.

To quantify association strength independently of sample size and table dimensions, Cramér's V is computed (denoted as A to prevent notation conflict with the rest-of-question text V):

$$A(C_i, C_j) = \sqrt{\frac{\chi^2}{n \cdot (\min(R_i, R_j) - 1)}}$$

where $A(C_i, C_j) \in [0, 1]$, with 0 indicating independence and 1 perfect association.

Phase 3: Association Filtering To eliminate weak or spurious relationships, a pair (C_i, C_j) is retained if and only if it satisfies a dual-threshold condition:

1. **Statistical Significance:** $p_{i,j} < \alpha \quad (\alpha = 0.005)$
2. **Effect Size:** $A_{i,j} \geq A_{\min} \quad (A_{\min} = 0.3)$

Surviving pairs are sorted by descending association strength to form a prioritized candidate queue:

$$\mathcal{E} = \{(C_i, C_j, A_{i,j}) \mid p_{i,j} < \alpha \wedge A_{i,j} \geq A_{\min}\}$$

Phase 4: Constrained Maximum Spanning Forest Partitioning Let $\mathcal{C} = \{C_1, \dots, C_M\}$ be the static, dataset-wide set of concepts corresponding to the table columns. Unlike natural-language datasets, $\mathcal{C}$ remains identical for every instance x.

The partitioning of $\mathcal{C}$ is modeled as finding a **Constrained Maximum Spanning Forest** on an undirected graph $G = (\mathcal{C}, \mathcal{E}, W)$, where edge weights are given by $w(C_i, C_j) = A(C_i, C_j)$. The partitioning is resolved via a modified Kruskal's algorithm constrained by a maximum component size ($n_{\max}$):

Algorithm 1 Constrained Greedy Partitioning

```
Initialize partition 𝒫 ← {{C} | C ∈ 𝒞}        ▷ All concepts start as singletons
Sort ℰ descending by weight A_{i,j}
for each edge e = (C_i, C_j, A_{i,j}) ∈ ℰ do
    Find sets P_a, P_b ∈ 𝒫 containing C_i and C_j
    if P_a ≠ P_b ∧ |P_a ∪ P_b| ≤ n_max then
        𝒫 ← (𝒫 \ {P_a, P_b}) ∪ {P_a ∪ P_b}      ▷ Merge groups
    end if
end for
return 𝒫
```

The resulting partition $\mathcal{P}$ serves as the tabular oracle output $S := \mathcal{P}$, which is computed once for the entire dataset. Downstream steps—joint counterfactual generation and Shapley-based redistribution (Section 3.2.3)—then proceed identically to the natural-language pipeline, with each block $P \in \mathcal{P}$ utilized as a correlation group $s \in S$.

3.2.2 Putting Joined Interventions into pratice

With an oracle $\mathcal{O}(x, C, n_{\max})$ implemented that can group concepts to a set S based either on an LLM's intuition or mathmatical evidence based on the dataset, it is now possible to intervene on the selected concepts. Because Matton et al.'s original metric already contain the proceedures needed to intervene on a concept, this code only needs to minimally adapted. Most notably, the counterfactual generator must be updated to be able to intervene on multiple concepts to enable the major rationale of measuring the joint effect those concepts have according to 3.1.2.

It is however not sufficient to mix all concepts in a correlation set $s \in S$ together to just intervene on them combined. Otherwise it would not be able to distinguish the *contributions* (more on that in 3.2.3) of indivitual concepts to the joint effect. But measuring the effect all concepts of the same set have together (the grand coalition) and then again the individual concepts on their own is not sufficient as well: If a set contains more than two concepts, a concept might for instance exhibit another behavior on its own or in the grand coalition compared to when its mixed with another concept as the example shows in **??**.

Example 3.2

Suppose an LLM answers whether a patient is likely suffering from bacterial meningitis. $S = \{\{C_1, C_2, C_3\}\}$.

- C_1: High Fever
- C_2: Neck Stiffness
- C_3: Positive lumbar puncture (test result)

The model behaves roughly as follows:

- Removing high fever alone barely changes the diagnosis because neck stiffness and lumbar puncture already provide sufficient evidence.
- However, removing high fever together with neck stiffness causes a much larger change than expected because the remaining lumbar puncture result alone is less convincing and could hint to multiple sclerosis for instance instead.
- Removing all three of course completely changes the prediction.

Thus the contribution of high fever depends strongly on whether neck stiffness is still present, even though this dependence is much weaker in the grand coalition. So the effects of intervening on C_1 are significantly if it happens within $\{C_1\}$ (alone), $\{C_1, C_2\}$ or the grand coalition $\{C_1, C_2, C_3\}$ are totally different.

Therefore its inevitable to investiage the models shift in prediciting for all $2^{|s|} - 1$ possible combinations for counterfactuals of a correlation set s. By using the slightly adapted procedures that where established by Matton et al. there is a counterfactual created for each of those combinations. After that the evaluation pipeline continues in collecting model responses $r = (y, e)$ with the prediction y and explanation e. There is no need to make any changing to the response collection step.

3.2.3 Adapting the CE: Introducing Shapley Values into the mix

Motivation

While now it's established, how to identify correlated concepts and how to intervene on them jointly after identification, it remains unsolved how to attribute the *causal effects* measured in the shift of prediction y.

Notably Matton et al.'s definition of *causal effect* (see 2.1) considers individual concepts. Having a CE value for multiple concepts thrown together would not be good: first, because it makes tracing the effect *individual concepts* have nearly impossible and therefore voiding one major strength of the metric and violating **D1**. The other other reason is that that it would require massive reworking of the following evaluation pipeline. Since the vectors for the *causal effect* and *explanation implied effect* need to be of the same length [3] for the *pearson correlation* to be calculated, by potentially reducing the length of the CE-vector, the EE-vector needs to be sized down too. This would introduce new errors and challenges and make it impossible to build on the foundation of the original metric. So that is no option, the CE-vector should still contain exactly $|C|$ items. But how to break down the joint effects into individual ones?

There are primitive approaches to this: for instance taking the effects of all 2^n coalitions a concept takes part in, from the concept intervened on to the the grand coalition and taking the average. But this is dangerous, the concepts own effect is heavily influenced by the impact other concepts in the coalition have. Fourtunatly, crafiting an own solution is not necessary, because Shapley values can be used from Game Theory that deal excactly with such scenario.

Shapley-Values

Shapley values where introduced by Lloyd Shapley in 1953 [41]. Shapley values—a concept from cooperative game theory—determine how to fairly distribute a total payout among players based on their individual contributions to the coalition. It calculates a player's allocation by evaluating their expected marginal contribution across all possible coalitions that the player would could form [41, 31, 33].

Assume there is a defined value function $v(T)$ for any subset $T \subseteq N$ where N is the set with all players. The value allotted to a player $m \in S$ is given by 3.3.

$$\phi_m(v) = \sum_{T \subseteq N \setminus \{m\}} \frac{|T|!\,(|N| - |T| - 1)!}{|N|!} \Big[v(T \cup \{m\}) - v(T)\Big] \tag{3.3}$$

where $\frac{|T|!\,(|N|-|T|-1)!}{|N|!}$ is the weighting factor and $\Big[v(T \cup \{m\}) - v(T)\Big]$ is the marginal contribution the player m gives.

Shapley proved that his solution is the only one that can satisfy 4 core axioms [41, 31, 33]:

- **Efficiency**: The sum of the individual values equals the total value created by the grand coalition ($\sum_{i \in N} \phi_i(v) = v(N)$).
- **Symmetry**: Identical players who contribute the exact same amount to any coalition receive identical payouts (if $v(T \cup \{i\}) = v(T \cup \{j\})$ for all $T \subseteq N \setminus \{i, j\}$, then $\phi_i(v) = \phi_j(v)$).
- **Dummy**: A player who adds zero value to every coalition receives zero payout (if $v(T \cup \{m\}) - v(T) = 0$ for all $T \subseteq N \setminus \{m\}$, then $\phi_m(v) = 0$).
- **Additivity**: If two independent games are played, a player's payout from the combined games is the sum of their payouts from each individual game ($\phi_m(v + w) = \phi_m(v) + \phi_m(w)$).

[3] The length of both is the number of concepts to investigate, so usually the $|C|$ of the concepts present in the question to investigate x.

It however must be noted that with exponentially growing larger coalitions, Shapley values get harder to compute too [33]. Since n_max got introduced within the counterfactual generation step in 3.2.1 this is not a major problem since the pratical bootleneck would be in gathering LLM responses for the exponential size of counterfactuals instead. In the case of shapley values there are approximation methods such as Monte Carlo sampling [31].

Shapley values have a history of usage within the field of *Explainable AI*. Most notably with *SHAP* where the contribution of an input feature to the final prediction is calculated [23]. But also in alayzing natural language explanations like in the *CC-Shap* framework [33], which already got introduced in this thesis in 2.2.2.

Implementing Shapley into CE definition

In the previous section, two questions were left unanswered: How to define the value function v and how to pratically integrate Shapley values into the new framework?

Its not far-fetched to fill in the equation 2.1 for estimating the *causal effect*. Doing so is a good decision to keep backwards compatibility (**D2**). However the equation is only defined for a single intervened concept until now. Moreover, in chapter 3.2.3 the decision was lined out to not change the method signature $CE(x, C_m)$, the CE should still be defined for a single concept. Therefore the solution is to dismember the equation in two parts: the part behind the equal sign will be the value function for the shapley game. The CE value then will be the payout alloted to the concept by the shapley function.

The value function must be adapted slightly to support joint interventions of multiple concepts. First, let T be a set of concepts and let $\mathbb{C}_i$ denote the set of all possible values (full domain) for a concept $C_i \in T$. The set of all reasonably possible joint values for T, denoted as $\mathbb{T}$, is the Cartesian product of these individual value sets $\mathbb{T} = \prod_{C_i \in T} \mathbb{C}_i$, with $t \in \mathbb{T}$ being the tuple of the original values of the concepts in T $t = (c_i)_{C_i \in T}$. Analog to Mattons notation is $\mathbb{T}' = \mathbb{T} \setminus \{t\}$ which is the set of possible counterfactual values - the multi-concept analog to $\mathbb{C}_{\gg}{}'$. Then the value function of the shapley game for set T is analogly to 2.1 defined as

$$v(T) = \frac{1}{|\mathbb{T}'|} \sum_{t' \in \mathbb{T}'} D_{\mathrm{KL}} \left(\mathbb{P}_M(Y \mid x_{t \to t'}) \parallel \mathbb{P}_M(Y \mid x) \right). \tag{3.4}$$

Now having defined the value funktion, the equation for estimating the *causal effect* can now be defined. As in the original metric the CE for the concept C_m and question x is in 3.5. Also note that $C_m \in s$ (the assingment to the set s happens based on whats described in 3.2.1.)

$$CE(x, C_m, s) = \sum_{T \subseteq s \setminus \{C_m\}} \frac{|T|!\,(|s| - |T| - 1)!}{|s|!} \Big[v(T \cup \{C_m\}) - v(T) \Big] \tag{3.5}$$

However to Matton et al. recall the faithfulness estimation via the pearson correlation coeficient 2.3 where the CE is called by $CE(x, C)$. Therefore a version without

s as a parameter has to be crafted but receives the set of concepts. The question-level CE-vector gets rid of C_m and s. Analoge to Matton et al. let C be the set of concepts present in the question x. Assume we have a helper where n_{max} for $\mathcal{O}$ now gets a constant value.

$$f_C(C_m) = s \in \mathcal{O}(x, C, 5) \quad \text{such that} \quad C_m \in s \tag{3.6}$$

Queriying the helper gives us the the set s where concept C_m belongs to.

$$CE(x, C) = \Big(CE\big(x, C_m, f_C(C_m)\big)\Big)_{m=1}^{|C|} \tag{3.7}$$

Therefore its posible to build a CE definition that matches Mattons original for the pearson correlation coeficient.

3.2.4 Normalize Shapley-CE by correction

Motivation / Problem

TLDR; Concepts that are grouped together in BBQ by default are between additive,independent. By default because of shapleys fairness [4] property, the sum of payouts equals the kldiv of the grand coalition. This punishes on average concepts that were grouped together.

Calculating t-score

Now called somethign else.

Calculating adjusted kldiv

3.2.5 Remaining evaluationn pipeline

Explain that now the alingment between EE and CE is still measured. However the CE definiton got changed to be based on the adjusted kldiv for each concept. Also explain why EE did not get reworked in any way.

3.3 Analysis

3.3.1 Differences between extention and original

With beatiful graphs

3.3.2 Check: are the Desirada fufilled?

With proof sketch for backwards compatibility (see presentation)

[4] not sure about that

3.3.3 How is the metric extention expected to influence results

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